

SMART INDIA HACKATHON 2026



Problem Statement ID – SIH26019
Problem Statement Title – National Digital
Platform for Research, Policy Innovation &
Evidence-Based Land Governance
Theme – Smart Automation
PS Category – Software
Team Name – Land Matrix
Team Leader – Harsh Shakyawar



THE PROBLEM

India has vast land-related data, research, policies and geospatial information, but they are fragmented across multiple departments, institutions and platforms. This makes it difficult for researchers and policymakers to discover, combine and use this information for evidence-based land governance.

Key Challenges

-  **Climate Change**
Rising risks to land, water and ecosystems.
-  **Rapid Urbanization**
Growing demand for infrastructure and land.
-  **Urban-Rural Land Transitions**
Pressure on agricultural and rural land.
-  **Increasing Land Disputes**
Conflicting records and ownership issues.
-  **Sustainable Land Use Planning**
Need for balanced and long term decisions.

OUR SOLUTION

LANDPOLICY AI

AI-powered Land Intelligence & Policy Decision Platform
A centralized digital knowledge and collaboration ecosystem that integrates land datasets, research, policies, geospatial information and analytical tools to enable evidence-based decision-making in land governance.

Core Features

-  **Centralized Knowledge Repository**
Datasets, research, policies, case studies, geospatial data.
-  **Geospatial Analysis**
Maps, land-use, infrastructure, environmental layers.
-  **AI-Powered Insights**
Understand datasets, analyze data, explain results.
-  **Policy Experimentation**
Compare scenarios, adjust priorities, test outcomes.
-  **Collaboration Ecosystem**
Government, researchers, universities, industry, citizens.
-  **Evidence-Based Recommendations**
With sources, methodology and missing data alerts.



AI POLICY COPILOT

Natural-language
policy questions →
actionable
analysis.



SMART SITE RANKING

Automatically
locations using
configurable
criteria.



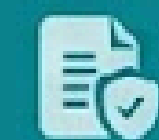
POLICY LAB

Test multiple
"What-if?" policy
scenarios.



EXPLAINABLE AI INSIGHTS

Explainable-language
Shows WHY each
recommendation
is generated.



EVIDENCE ENGINE

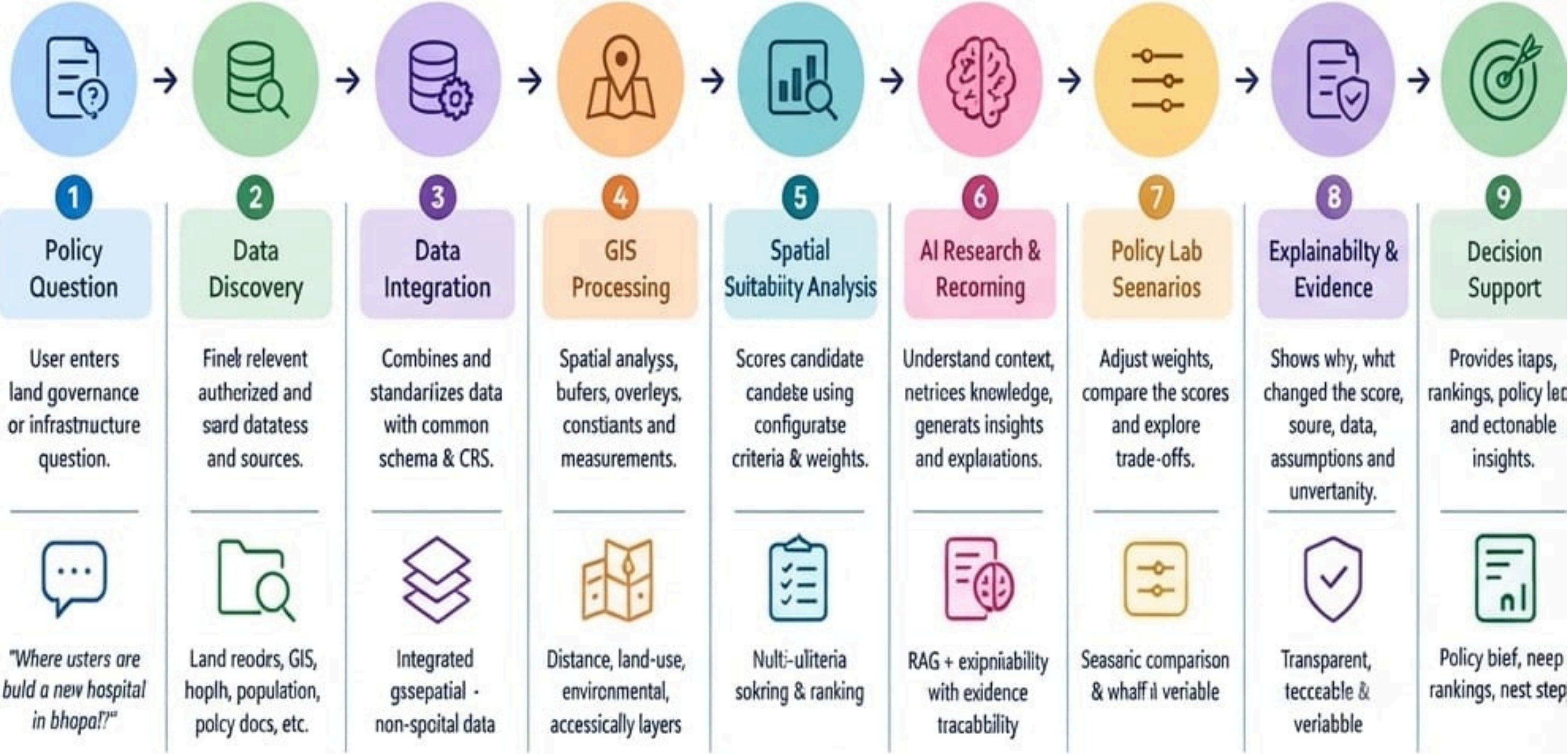
Links decision to data,
research and
methodology.



NATIONAL SCALE

Designed to expand
from one city to a
national ecosystem.

LANDPOLICY AI – END TO END WORKFLOW



Layer

Technology

Frontend

HTMLS, CSS3,
JavaScript,
Leaflet

Backend

Python,
FastAPI / Flask

AI / NLP

LLM / RAG,
Document Parsing,
Explainability

GIS & Spatial

PostGIS, GeoServer,
GeoPandas, Rasterio

Data

JSON / GeoJSON,
Authorized & Open
Government Datasets

Deployment

Docker,
Cloud / Data Centre

"Where users are
build a new hospital
in bhopal?"

Land records, GIS,
population, policy docs, etc.

Integrated
geospatial -
non-spatial data

Distance, land-use,
environmental,
accessibility layers

Multi-criteria
scoring & ranking

RAG + explainability
with evidence
traceability

Sensitivity comparison
& what-if variable

Transparent,
reliable &
variable

Policy brief, next
rankings, next step

FEASIBILITY AND VIABILITY

The platform is designed as an intelligence layer over existing land-governance information and systems.

FEASIBLE	CHALLENGES	MITIGATION
Uses existing digital documents, datasets and GIS evidence.	Different formats and inconsistent terminology.	Normalize and tag evidence before linking.
Can connect with outputs from existing land systems.	Source quality, completeness and update cycles.	Retain source context and traceable references.
Modular pipeline allows sources to be added progressively.	Complex links between evidence, geography and policy.	Use validation + human review for critical outputs.

Viability principle: start with high-value evidence sources, then expand the graph.

Makeland-governanceevidenceeeasier to discover, connect, compare and use.



Social • Better access to evidence Economic • More efficient research & policy analysis Governance • Stronger evidence linkage

Referencebasefortheproposedevidenceandland-governance intelligence layer.

DILRMP

Digital India Land Records Modernization Programme — land-record modernization context.

NAKSHA

National Geospatial Knowledge-based Land Survey of Urban Habitations — spatial/urban land context.

SVAMITVA

Survey of Villages and Mapping with Improved Technology in Village Areas — rural property mapping context.

